Lab 15

WSO2 Identity Server as the Key Manager

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **<APIM_HOST>** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **<HTTPS_PORT>** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal, Admin Portal, Management Console, token endpoint).
- **<GW_HTTPS_PORT> / <GW_HTTP_PORT>** – API Gateway ports: 8243 (HTTPS) and 8280 (HTTP) plus any port offset.
- **<APIM_HOME>** – the folder where API Manager is extracted.

The servers use a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue.

This lab also uses these placeholders for Identity Server:

- **<IS_HOST>** – host name or IP address of the Identity Server (localhost when it runs on the same machine as API Manager).
- **<IS_HTTPS_PORT>** – Identity Server HTTPS port: 9443 plus its port offset. With offset 1, as in this lab, it is 9444.
- **<IS_HOME>** – the folder where Identity Server is extracted.
- **<JDK25_DIR>** – the folder where JDK 25 is installed (Windows only, see step 3).

Training Objective

Configure WSO2 Identity Server 7.3 as a Key Manager for WSO2 API Manager 4.7.0, generate application keys and access tokens from Identity Server, and use those tokens to call an API through the API Manager gateway.

Business Scenario

PizzaShack wants to use WSO2 Identity Server as the Key Manager of API Manager, so that authentication and authorization for its APIs are controlled by Identity Server.

High-Level Steps

- Install JDK 25 and download Identity Server
- Configure Identity Server (port offset, Key Manager settings, event handler)
- Exchange the server certificates, then start Identity Server and API Manager with JDK 25
- Add Identity Server as a Key Manager in the Admin Portal
- Generate keys and a token from Identity Server and call an API

Before You Start

This lab assumes API Manager runs with no port offset (HTTPS port 9443) and Identity Server runs on the same machine with port offset 1 (HTTPS port 9444). Stop any other server that uses port

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9444. You also need the file wso2is.notification.event.handlers-2.1.3.jar from the Day 3 lab files and the PizzaShack API from Day 1.

JDK 25: in this lab both Identity Server 7.3.0 and API Manager 4.7.0 run on JDK 25. Part 1 shows how to install it and make it the Java that the servers use.

Part 1 – Install JDK 25 and Download Identity Server

1. Check which Java version the terminal uses. If it already shows version 25, go to step 4.

```
java -version
```

2. Install JDK 25 (Eclipse Temurin 25):

macOS (Homebrew)

```
brew install --cask temurin@25
```

Windows (Command Prompt)

```
winget install EclipseAdoptium.Temurin.25.JDK
```

On macOS, Homebrew asks for your Mac password. On Windows, accept the installer prompt. On Linux, install a JDK 25 package with your package manager.

3. Make JDK 25 the Java used by the servers. Do this in every terminal you start a server from in this lab:

macOS

```
export JAVA_HOME=$(/usr/libexec/java_home -v 25)
export PATH="$JAVA_HOME/bin:$PATH"
```

To make it permanent on macOS, add the same two lines to `~/.zshrc`.

On Windows, the installer sets JAVA_HOME and PATH. Close the Command Prompt and open a new one, then run `echo %JAVA_HOME%`. If it shows another JDK, set it to the JDK 25 folder and open a new Command Prompt again:

Windows (Command Prompt)

```
setx JAVA_HOME "<JDK25_DIR>"
```

Example – Windows (Command Prompt):

```
setx JAVA_HOME "C:\Program Files\Eclipse Adoptium\jdk-25.0.5.8-hotspot"
```

Check the result: `java -version` must show `openjdk version "25`.

4. Download WSO2 Identity Server 7.3.0 from the WSO2 downloads page (fill in the form if it asks for your details) and extract the zip. The extracted folder (for example `wso2is-7.3.0`) is <IS_HOME>.

Download page

```
https://wso2.com/products/downloads/?product=wso2is
```

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Part 2 – Configure Identity Server

5. Open `<IS_HOME>/repository/conf/deployment.toml` and add `offset = 1` under the existing `[server]` section:

```
[server]
hostname = "localhost"
node_ip = "127.0.0.1"
base_path = "https://$ref{server.hostname}:${carbon.management.port}"
offset = 1
```

6. Add the following at the end of the same file. `notification_endpoint` is the API Manager internal notification URL; Identity Server calls it to tell API Manager when a token is revoked.

```
[oauth]
authorize_all_scopes = true

[[resource.access_control]]
context = "(.*)/scim2/Me"
secure = true
http_method = "GET"
cross_tenant = true
permissions = []
scopes = []

[[event_listener]]
id = "token_revocation"
type = "org.wso2.carbon.identity.core.handler.AbstractIdentityHandler"
name = "org.wso2.is.notification.ApimOauthEventInterceptor"
order = 1

[event_listener.properties]
notification_endpoint = "https://<APIM_HOST>:<HTTPS_PORT>/internal/data/v1/notify"
username = "${admin.username}"
password = "${admin.password}"
"header.X-WSO2-KEY-MANAGER" = "WSO2-IS"
```

Example:

```
[oauth]
authorize_all_scopes = true

[[resource.access_control]]
context = "(.*)/scim2/Me"
secure = true
http_method = "GET"
cross_tenant = true
permissions = []
scopes = []

[[event_listener]]
id = "token_revocation"
type = "org.wso2.carbon.identity.core.handler.AbstractIdentityHandler"
name = "org.wso2.is.notification.ApimOauthEventInterceptor"
order = 1

[event_listener.properties]
notification_endpoint = "https://localhost:9443/internal/data/v1/notify"
username = "${admin.username}"
password = "${admin.password}"
"header.X-WSO2-KEY-MANAGER" = "WSO2-IS"
```

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7. Copy `wso2is.notification.event.handlers-2.1.3.jar` to `<IS_HOME>/repository/components/dropins`. Run the command from the folder that contains the jar:

macOS / Linux

```
cp wso2is.notification.event.handlers-2.1.3.jar \
  <IS_HOME>/repository/components/dropins/
```

Example – macOS / Linux:

```
cp wso2is.notification.event.handlers-2.1.3.jar \
  ~/wso2is-7.3.0/repository/components/dropins/
```

Windows (Command Prompt)

```
copy wso2is.notification.event.handlers-2.1.3.jar ^
  <IS_HOME>\repository\components\dropins\
```

Example – Windows (Command Prompt):

```
copy wso2is.notification.event.handlers-2.1.3.jar ^
  C:\wso2is-7.3.0\repository\components\dropins\
```

Part 3 – Exchange the Certificates and Start the Servers

Each server must trust the other. Export the public certificate of each server and import it into the truststore of the other. Identity Server 7.3 uses PKCS12 stores (.p12); API Manager 4.7.0 uses JKS stores (.jks). The default store password of both is wso2carbon.

8. Run these commands from any working folder. keytool comes with the JDK installed in Part 1:

macOS / Linux

```
keytool -exportcert -rfc -alias wso2carbon -file is.crt \
  -keystore <IS_HOME>/repository/resources/security/wso2carbon.p12 \
  -storepass wso2carbon
keytool -exportcert -rfc -alias wso2carbon -file apim.crt \
  -keystore <APIM_HOME>/repository/resources/security/wso2carbon.jks \
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2is7 -file is.crt \
  -keystore <APIM_HOME>/repository/resources/security/client-truststore.jks \
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2apim -file apim.crt \
  -keystore <IS_HOME>/repository/resources/security/client-truststore.p12 \
  -storepass wso2carbon
```

Example – macOS / Linux:

```
keytool -exportcert -rfc -alias wso2carbon -file is.crt \
  -keystore ~/wso2is-7.3.0/repository/resources/security/wso2carbon.p12 \
  -storepass wso2carbon
keytool -exportcert -rfc -alias wso2carbon -file apim.crt \
  -keystore ~/wso2am-4.7.0/repository/resources/security/wso2carbon.jks \
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2is7 -file is.crt \
  -keystore ~/wso2am-4.7.0/repository/resources/security/client-truststore.jks \
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2apim -file apim.crt \
  -keystore ~/wso2is-7.3.0/repository/resources/security/client-truststore.p12 \
  -storepass wso2carbon
```

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Windows (Command Prompt)

```
keytool -exportcert -rfc -alias wso2carbon -file is.crt ^
  -keystore <IS_HOME>\repository\resources\security\wso2carbon.p12 ^
  -storepass wso2carbon
keytool -exportcert -rfc -alias wso2carbon -file apim.crt ^
  -keystore <APIM_HOME>\repository\resources\security\wso2carbon.jks ^
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2is7 -file is.crt ^
  -keystore <APIM_HOME>\repository\resources\security\client-truststore.jks ^
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2apim -file apim.crt ^
  -keystore <IS_HOME>\repository\resources\security\client-truststore.p12 ^
  -storepass wso2carbon
```

Example – Windows (Command Prompt):

```
keytool -exportcert -rfc -alias wso2carbon -file is.crt ^
  -keystore C:\wso2is-7.3.0\repository\resources\security\wso2carbon.p12 ^
  -storepass wso2carbon
keytool -exportcert -rfc -alias wso2carbon -file apim.crt ^
  -keystore C:\wso2am-4.7.0\repository\resources\security\wso2carbon.jks ^
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2is7 -file is.crt ^
  -keystore C:\wso2am-4.7.0\repository\resources\security\client-truststore.jks ^
  -storepass wso2carbon
keytool -importcert -noprompt -alias wso2apim -file apim.crt ^
  -keystore C:\wso2is-7.3.0\repository\resources\security\client-truststore.p12 ^
  -storepass wso2carbon
```

Each import prints **Certificate was added to keystore**.

9. Start Identity Server with JDK 25 and wait for **WSO2 Carbon started** in the console. Use a terminal where `java -version` shows 25 (step 3):

macOS / Linux

```
cd <IS_HOME>/bin
sh wso2server.sh
```

Example – macOS / Linux:

```
cd ~/wso2is-7.3.0/bin
sh wso2server.sh
```

Windows (Command Prompt)

```
cd <IS_HOME>\bin
wso2server.bat
```

Example – Windows (Command Prompt):

```
cd C:\wso2is-7.3.0\bin
wso2server.bat
```

The start log shows the Java in use, for example `Java Version : 25.0.5`. If it shows another version, stop the server (Ctrl+C), repeat step 3 in that terminal and start again. If the log shows `Address already in use`, the port offset from step 5 is missing. The Identity Server Console is then available at:

```
https://<IS_HOST>:<IS_HTTPS_PORT>/console
```

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Example:

```
https://localhost:9444/console
```

10. Restart API Manager with JDK 25, so that it runs on the same Java and loads the Identity Server certificate imported in step 8. Stop it (Ctrl+C in its terminal), then start it again from a terminal where `java -version` shows 25:

macOS / Linux

```
cd <APIM_HOME>/bin
sh api-manager.sh
```

Example – macOS / Linux:

```
cd ~/wso2am-4.7.0/bin
sh api-manager.sh
```

Windows (Command Prompt)

```
cd <APIM_HOME>\bin
api-manager.bat
```

Example – Windows (Command Prompt):

```
cd C:\wso2am-4.7.0\bin
api-manager.bat
```

Check that the API Manager log also shows `Java Version : 25` at startup.

Part 4 – Add Identity Server as a Key Manager

11. Log in to the Admin Portal as admin, click **Key Managers** and then **Add Key Manager**:

```
https://<APIM_HOST>:<HTTPS_PORT>/admin
```

Example:

```
https://localhost:9443/admin
```

12. Enter Name **WSO2IS7**, Display Name **WSO2 IS 7** and select the Key Manager Type **WSO2 Identity Server 7**.

13. In **Well-known URL** enter the URL below and click **Import**. This fills the issuer, the endpoints, the grant types and the JWKS certificate URL.

```
https://<IS_HOST>:<IS_HTTPS_PORT>/oauth2/token/.well-known/openid-configuration
```

Example:

```
https://localhost:9444/oauth2/token/.well-known/openid-configuration
```

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14. Correct the two fields that Import does not set as needed:

Field	Value	Why
UserInfo Endpoint	`https://<IS_HOST>:<IS_HTTPS_PORT>/scim2/Me`	Import fills /oauth2/userInfo
Scope Management Endpoint	`https://<IS_HOST>:<IS_HTTPS_PORT>/api/identity/oauth2/v1.0/scopes`	Import leaves it empty

15. Check the other values. All endpoints start with https://<IS_HOST>:<IS_HTTPS_PORT>:

Field	Path
Issuer, Token Endpoint, Display Token Endpoint	`/oauth2/token`
Client Registration Endpoint	`/api/identity/oauth2/dcr/v1.1/register`
Introspection Endpoint	`/oauth2/introspect`
Revoke Endpoint, Display Revoke Endpoint	`/oauth2/revoke`
Authorize Endpoint	`/oauth2/authorize`
Certificates: JWKS URL	`/oauth2/jwks`

16. Under **Connector Configurations**, enter the values below. Leave **Create roles in WSO2 Identity Server 7** switched off.

Field	Value
Authentication Type	Basic Authentication
Username / Password	The Identity Server admin user (default admin / admin)
WSO2 Identity Server 7 API Resource Management Endpoint	`https://<IS_HOST>:<IS_HTTPS_PORT>/api/server/v1/api-resources`
WSO2 Identity Server 7 Roles Endpoint	`https://<IS_HOST>:<IS_HTTPS_PORT>/scim2/v2/Roles`

Example:

```
WSO2 Identity Server 7 API Resource Management Endpoint:
https://localhost:9444/api/server/v1/api-resources

WSO2 Identity Server 7 Roles Endpoint:
https://localhost:9444/scim2/v2/Roles
```

Use the Identity Server port, not 9443: both fields are required, and their hints show port 9443 (for example `https://localhost:9443/scim2/v2/Roles`). In this lab Identity Server runs with port offset 1, so the endpoints use port 9444. Port 9443 is API Manager.

17. Click **Add**. WSO2IS7 now appears in the Key Managers list.

Part 5 – Test the Key Manager

18. Log in to the Developer Portal and open an application that is subscribed to the PizzaShack API (or create one and subscribe it):

```
https://<APIM_HOST>:<HTTPS_PORT>/devportal
```

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Example:

```
https://localhost:9443/devportal
```

19. Open **Production Keys**. There is now a tab for each Key Manager: select **WSO2 IS 7**, tick the **Client Credentials** grant type and click **Generate Keys**. The OAuth application is created in Identity Server. Copy the consumer key and secret.

20. Click **Generate Access Token** and copy the token. Alternatively, get the token directly from Identity Server:

    macOS / Linux

    ```
    curl -k -u "<CONSUMER_KEY>:<CONSUMER_SECRET>" \
      -d "grant_type=client_credentials" \
      "https://<IS_HOST>:<IS_HTTPS_PORT>/oauth2/token"
    ```

    Example – macOS / Linux:

    ```
    curl -k -u "Xb3fK9qLm2:p7TzR4wYc1" \
      -d "grant_type=client_credentials" \
      "https://localhost:9444/oauth2/token"
    ```

    Windows (Command Prompt)

    ```
    curl -k -u "<CONSUMER_KEY>:<CONSUMER_SECRET>" ^
      -d "grant_type=client_credentials" ^
      "https://<IS_HOST>:<IS_HTTPS_PORT>/oauth2/token"
    ```

    Example – Windows (Command Prompt):

    ```
    curl -k -u "Xb3fK9qLm2:p7TzR4wYc1" ^
      -d "grant_type=client_credentials" ^
      "https://localhost:9444/oauth2/token"
    ```

21. Call the API through the API Manager gateway with the Identity Server token:

    macOS / Linux

    ```
    curl -k -H "Authorization: Bearer <IS_ACCESS_TOKEN>" \
      "https://<APIM_HOST>:<GW_HTTPS_PORT>/pizzashack/1.0.0/menu"
    ```

    Example – macOS / Linux:

    ```
    curl -k -H "Authorization: Bearer eyJ4NXQiOi.eyJzdWIi.ZXhhbXBsZQ" \
      "https://localhost:8243/pizzashack/1.0.0/menu"
    ```

    Windows (Command Prompt)

    ```
    curl -k -H "Authorization: Bearer <IS_ACCESS_TOKEN>" ^
      "https://<APIM_HOST>:<GW_HTTPS_PORT>/pizzashack/1.0.0/menu"
    ```

    Example – Windows (Command Prompt):

    ```
    curl -k -H "Authorization: Bearer eyJ4NXQiOi.eyJzdWIi.ZXhhbXBsZQ" ^
      "https://localhost:8243/pizzashack/1.0.0/menu"
    ```

    The gateway validates the Identity Server token (a JWT) and returns HTTP 200 with the menu.

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Expected Outcome

Identity Server 7.3 acts as a Key Manager for API Manager 4.7.0, both running on JDK 25: applications get their keys and access tokens from Identity Server, and the API Manager gateway accepts those tokens.

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